



# Dual-aspect synergistic outlier detection with structural deviation and attribute rarity

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## ARTICLE INFO

### Keywords:

Outlier detection  
Granular computing  
Formal concept analysis  
Core concept

## ABSTRACT

Outlier detection is crucial in domains such as financial fraud detection, cybersecurity and industrial fault diagnosis. Although formal concept analysis (FCA) offers an interpretable framework for outlier detection by measuring concept deviation, few existing methods only consider object membership in granular concepts and neglect attribute-level information, resulting in an incomplete characterization of outlier behaviors. To address these limitations, this paper proposes a dual-aspect synergistic outlier detection method based on FCA. The framework first measures structural deviation in combination from both object and attribute perspectives. Additionally, a noise-robust core concept extraction strategy filters incidental concepts, emphasizes dominant data patterns and reduces computational complexity. Finally, the model integrates conceptual structural deviation with statistically significant attribute-level rarity computed from self-information to generate a unified outlier score. Experimental evaluations on 16 benchmark datasets demonstrate that our method outperforms eight other recently proposed representative models. Moreover, it yields better interpretability by leveraging the duality of concept extents and intents, which is further illustrated through a concrete case study.

## 1. Introduction

Anomaly detection aims to identify data instances that deviate significantly from the majority of observations [1]. It is vital for real-world applications such as detecting financial fraud [2], network intrusions [3], industrial faults [4], and medical diseases [5]. Timely anomaly detection helps avert severe losses, enhance system reliability, and even safeguard human lives. Given its broad applicability and practical importance, anomaly detection has attracted extensive research attention across multiple disciplines [6].

As a subtype within the broader anomaly detection framework, outlier detection specifically refers to the statistical identification of individual data points that deviate significantly from the underlying data distribution [7]. Researchers have developed a wide variety of outlier detection methods, which can be primarily categorized into statistical-based [8], distance-based [9], density-based [10], and clustering-based approaches [11]. A common limitation of these methods is their reliance on geometric computations in continuous spaces, often rendering them unsuitable for categorical or mixed-type datasets [12]. Another inherent drawback is that most outlier detection methods yield only

uninterpretable numerical scores, which barely reveal the specific attributes causing deviations or the reasons why an instance is identified as anomalous. This lack of interpretability has severely hindered the application of these methods in high-stakes domains such as network security, finance and industrial systems [13].

Granular computing (GrC) [14] provides paradigms that mimic human cognitive processes by organizing data into coherent “information granules”. Within this framework, rough set theory (RST) [15] serves as a core methodology, which models uncertainty through equivalence classes and can be effectively applied to diverse types of data. However, traditional RST-based approaches typically partition the data space into a flat and non-hierarchical collection of granules, potentially limiting their ability to capture the complex multi-level relationships among instances [16]. In contrast, formal concept analysis [17–19], a knowledge-driven framework, constructs an inherently interpretable structure by representing object-attribute dual relationships in a hierarchical concept lattice and performs well in various data mining tasks.

Recently, Hu et al. proposed two FCA-based methods for outlier detection, namely GCOD [20] and TIEOD [21]. These approaches identify

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structural relationships among granular concepts by measuring the differences in their extents, enabling the detection of anomalous granular concepts in the data space. Outliers are subsequently determined based on conceptual anomalies. Granular concepts are capable of reflecting local density information while achieving clustering effects simultaneously. The object-attribute duality of anomalous concepts in the sense of category theory provides interpretability for the outlier detection task. However, existing FCA-based outlier detection frameworks have two key limitations. First, they measure the distances between concepts solely based on object membership, without considering geometric positional differences of granular concepts in the data space. This hinders the global and holistic analysis of outliers. Second, these models fail to incorporate attribute-level information, resulting in a lack of quantification of the intrinsic distribution properties.

To address the above limitations, it is necessary to develop an FCA-based outlier detection approach that integrates concept membership relationships, geometric positions, and attribute distribution characteristics. Motivated by this, we propose a dual-aspect synergistic outlier detection method with structural deviation and attribute rarity (DASOD). This method couples two complementary aspects: the structural deviation identified by core concepts and the statistical rarity measurements of attributes. Instances that exhibit both structural deviation and statistical rarity are thereby identified as outliers. The main contributions of this paper are as follows.

- The framework evaluates structural deviation through two adjacent dimensions (concept extent and intent) synchronously. For extent, deviation reflects object membership composition differences across granular concepts; for intent, it considers geometric distances among stable attributes of granular concepts. Integrating these two measurements achieves comprehensive structural deviation characterization.
- High-support granular concepts representing dominant, stable data patterns are selected as core concepts. Structural deviation of granular concepts is only measured against core concepts, thus significantly reducing concept scale and filtering out noise-susceptible incidental concepts.
- From an information theory perspective, we measure the statistical rarity of attributes by calculating the self-information of each object's attribute values, where objects with rare attribute combinations are assigned higher statistical scores. At last, the outlier degree from conceptual structural deviation and statistical attribute rarity is coupled via multiplication for each instance.

## 2. Related work

This section summarizes the research landscape relevant to our work, which is organized into four parts covering classical outlier detection methods, RST-based approaches, formal concept analysis foundations, and FCA-based outlier detection.

Unsupervised outlier detection methods can be roughly categorized into several paradigms. Statistical-based approaches [8,22] identified outliers in low-probability regions, but their reliance on strong distributional assumptions often limits their practical applicability. To address this, Li et al. proposed non-parametric methods COPOD [23] and ECOD [24]. By utilizing copula functions and empirical cumulative distribution functions, these methods captured tail probabilities without assuming specific distributions. Proximity-based methods, exemplified by KNN [25] and LOF [10], quantified outlier degree through neighborhood distances or local density ratios. Despite their intuitiveness, they are often sensitive to hyper-parameters and the curse of dimensionality. Clustering-based methods [11] identified outliers through structural deviations, whereas reconstruction-based methods such as PCA [26] and autoencoders [27] relied on large reconstruction errors. Deep learning-based anomaly detection methods such as Deep SVDD [28] have emerged as a dominant paradigm for modeling complex nonlinear

data patterns. Beyond traditional paradigms, some recent studies integrated entropy, density or deep learning techniques for more robust detection. Zhang et al. [29] proposed the EDROD method, which combines local Shannon entropy with global density estimation to simultaneously detect point and cluster outliers. Goodge et al. [30] introduced LUNAR, which integrates conventional local outlier strategies into graph neural networks to learn neighborhood structures and capture nonlinear patterns adaptively. Xu et al. [31] proposed DIF, which leverages neural representations to implement nonlinear feature partitioning. These methods primarily operate on numerical proximity or reconstruction errors, offering limited insights into the conceptual or structural semantics underlying anomalous behavior.

Granular computing facilitates outlier detection by evaluating how objects interact with information granules. Early RST-based methods, such as those by Jiang et al. [32] and Shaari et al. [33], identified outliers as objects residing in boundary regions that increase the uncertainty of the knowledge base. To handle continuous and hybrid data, fuzzy-rough sets (FRS) and neighborhood-based granulation have been extensively explored by Yuan et al. [34]. Recently, Su et al. [35] proposed GBDO, which utilizes fuzzy relations between granular balls to measure local reachability similarity density, thereby solving the noise sensitivity issue common in single-granularity methods. They further proposed MGBOD [36], which fuses multi-granularity fuzzy information to capture intricate data structures and address the limitations of fine-grained representations. Yang et al. [37] proposed 3W-SGBD by integrating shadowed set theory with the principle of justifiable granularity, effectively solving the problem of boundary fuzziness through a dual-entropy-driven mechanism. However, most rough set-based approaches rely on flat or loosely coupled granules, which limits their ability to model hierarchical conceptual relationships among data objects.

FCA provides a mathematically rigorous lattice-theoretic framework for deriving hierarchical concepts from binary relations. Despite its strong interpretability and success in domains such as rule extraction [38] and machine learning [39], the exponential growth of the concept lattice often hinders its scalability on high-dimensional data. To alleviate this bottleneck, researchers have explored attribute reduction [40] and concept pruning [41]. In parallel, granular computing perspectives have been incorporated into FCA to provide multi-granular foundations for concept learning and knowledge reduction [42,43]. This challenge has motivated research toward representative, reduced, or structurally significant granular concepts, rather than exhaustive construction of the entire concept lattice.

Driven by the demand for structural interpretability, researchers have recently explored FCA as a framework for unsupervised outlier detection. Hu et al. introduced a granular concept-based outlier detection model (GCOD) [20] and extended their ideas into three-way formal concepts with TIEOD [21]. Liu et al. [44] proposed an entropy-based method (OEOD) specifically for nominal data. However, these methods focus on a single conceptual perspective or local information only, while overlooking the collaborative consideration of both objects and attributes as well as the global measurement of anomalies. To bridge this gap, we propose the Dual-Aspect Synergistic Outlier Detection (DASOD) framework, which enhances synergy between different conceptual aspects for more robust detection.

## 3. Preliminaries

To facilitate the subsequent discussion, we briefly revisit the necessary background knowledge while referring interested readers to [9,17] for a more comprehensive treatment.

### 3.1. Outlier detection

According to Hawkins [45], the intuitive definition of an outlier would be “an observation which deviates so much from other observations as to arouse suspicions that it was generated by a different mechanism”. For data mining, Knorr and Ng [9] give the following distance-based operational definition of outliers.

**Definition 1 ([9]).** Given a dataset  $U$ , a distance metric  $d$  on  $U$  and a threshold  $\delta \geq 0$ , for any  $x \in U$ , its  $\delta$ -neighborhood fraction  $\theta(x)$  is defined as:

$$\theta(x) = \frac{|\{y \in U \mid \text{dist}(x, y) \leq \delta\}|}{|U|}. \quad (1)$$

The object  $x$  is identified as a distance-based outlier if its neighborhood density falls below the specified threshold, i.e.,  $\theta(x) \leq \eta$ , where  $\eta \in [0, 1]$  denotes a fraction threshold.

### 3.2. Formal concept analysis

**Definition 2 ([17]).** Let  $\mathcal{K} = (U, A, I)$  be a formal context, where  $U$  and  $A$  are finite nonempty sets of objects and attributes, respectively and  $I \subseteq U \times A$  denotes a binary relation. The notation  $(x, a) \in I$  indicates that object  $x$  possesses attribute  $a$ .

A formal context is said to be regular if it satisfies the following conditions, ensuring structural non-triviality:

- (1) For any  $x \in U$ , there exist  $a_1, a_2 \in A$  such that  $(x, a_1) \in I$  and  $(x, a_2) \notin I$ ;
- (2) For any  $a \in A$ , there exist  $x_1, x_2 \in U$  such that  $(x_1, a) \in I$  and  $(x_2, a) \notin I$ .

Throughout this paper, we assume that all formal contexts are regular unless stated otherwise.

**Definition 3 ([17]).** Let  $\mathcal{K} = (U, A, I)$  be a formal context. For any object subset  $X \subseteq U$  and attribute subset  $B \subseteq A$ , the pair of operators  $\uparrow$  and  $\downarrow$  is defined as follows:

$$X^\uparrow = \{a \in A \mid \forall x \in X, (x, a) \in I\}, \quad (2)$$

$$B^\downarrow = \{x \in U \mid \forall a \in B, (x, a) \in I\}. \quad (3)$$

In particular,  $X^\uparrow$  represents the set of common attributes shared by all objects in  $X$ , while  $B^\downarrow$  denotes the set of objects that possess every attribute in  $B$ .

The pair of operators defined in Definition 3 constitutes a Galois connection between the power sets  $\mathcal{P}(U)$  and  $\mathcal{P}(A)$ . We refer the reader to [19] for details.

**Definition 4 ([17]).** Let  $\mathcal{K} = (U, A, I)$  be a formal context. A pair  $(X, B)$  with  $X \subseteq U$  and  $B \subseteq A$  is termed a formal concept (or simply a concept) if and only if it satisfies the closure property:  $X^\uparrow = B$  and  $B^\downarrow = X$ . In particular,  $X$  and  $B$  are called the extent and intent of the concept  $(X, B)$ , respectively.

The collection of all formal concepts derived from  $\mathcal{K}$  forms a complete lattice, denoted by  $\mathcal{L}$ . For two concepts  $C_1 = (X_1, B_1), C_2 = (X_2, B_2) \in \mathcal{L}$ , a partial order relation  $\leq$ , representing the concept hierarchy, is defined as:  $(X_1, B_1) \leq (X_2, B_2) \iff X_1 \subseteq X_2 \iff B_2 \subseteq B_1$ . The infimum and supremum operations on  $\mathcal{L}$  are given by:

$$(X_1, B_1) \wedge (X_2, B_2) = (X_1 \cap X_2, (B_1 \cup B_2)^\downarrow), \quad (4)$$

$$(X_1, B_1) \vee (X_2, B_2) = ((X_2 \cup X_2)^\downarrow, B_1 \cap B_2). \quad (5)$$

## 4. Our proposed approach

This section presents the proposed DASOD algorithm. The DASOD algorithm incorporates the following designs: (1) dual-dimensional structural deviation measurement, (2) core concept screening mechanism, and (3) coupling of global attribute rarity based on self-information statistical score and local structural deviation. The specific procedures for model construction and outlier scoring are detailed in Algorithms 1 and 2, respectively.

### 4.1. Model construction

**Definition 5 ([42]).** Let  $\mathcal{K} = (U, A, I)$  be a formal context.  $\mathcal{L}_{og}(\mathcal{K}) = \{(x^\downarrow, x^\uparrow) \mid x \in U\}$  and  $\mathcal{L}_{ag}(\mathcal{K}) = \{(a^\downarrow, a^\uparrow) \mid a \in A\}$  are referred to as the sets of object granular concepts and the attribute granular concepts, respectively. As a subset of  $\mathcal{L}$ ,  $\mathcal{L}_g = \mathcal{L}_{og}(\mathcal{K}) \cup \mathcal{L}_{ag}(\mathcal{K})$  denotes the set of granular concepts.

**Proposition 1.** Let  $\mathcal{K} = (U, A, I)$  be a formal context. For any formal concept  $(X, B)$  in the concept lattice  $\mathcal{L}$ , the following result holds:

$$(X, B) = \bigvee_{x \in X} (x^\downarrow, x^\uparrow) = \bigwedge_{a \in B} (a^\downarrow, a^\uparrow). \quad (6)$$

It is not difficult to prove the above proposition based on the definition of formal concepts. From the perspective of order relations, this indicates that object granular concepts and attribute granular concepts respectively serve as atoms and co-atoms of the concept lattice  $\mathcal{L}$  and thus can reconstruct the entire concept lattice. Granular concepts provide an effective means for modeling the object-attribute structure in formal contexts and are often adopted to replace the entire concept space in practical concept-cognitive learning tasks.

**Definition 6 ([20]).** Assume  $\mathcal{K} = (U, A, I)$  is a formal context. For any  $x, y \in U$ , a granular concept-based distinction degree ( $GCD$ ) between  $x$  and  $y$  is given by:

$$GCD(x, y) = |\{(X, B) \in \mathcal{L}_g \mid x \in X \oplus y \in X\}|. \quad (7)$$

Intuitively,  $GCD$  can be interpreted as the Hamming distance [10] between the membership indicator vectors of objects over all granular concepts  $\mathcal{L}_g$ . Formula (7) uses the membership relation to calculate the number of granular concepts that can distinguish two objects, thereby determining the degree of distinction between them.

**Proposition 2 ([20]).** Assume  $\mathcal{K} = (U, A, I)$  is a formal context. The  $GCD$  defined above is a metric on  $U$ .

By means of  $GCD$ , the extent-based distinction degree between granular concepts can be defined.

**Definition 7 ([20]).** Let  $\mathcal{K} = (U, A, I)$  be a formal context and  $\mathcal{L}_g$  denote the set of granular concepts derived from  $\mathcal{K}$ . For any two concepts  $C_1 = (X_1, B_1), C_2 = (X_2, B_2) \in \mathcal{L}_g$ , the extent-based distinction degree between them is defined as:

$$M(C_1, C_2) = \begin{cases} \frac{\sum_{x \in X_1} \sum_{y \in X_2} GCD(x, y)}{|X_1| |X_2|}, & C_1 \neq C_2, \\ 0, & C_1 = C_2. \end{cases} \quad (8)$$

**Proposition 3 ([20]).** The extent-based distinction degree  $M$  defined above is a metric on  $\mathcal{L}_g$ .

In the framework of FCA, equal-width discretization of features is commonly adopted to transform a dataset into a formal context, with the number of discretization intervals described as granularity. Considering that granular concepts differ not only in their extents, but also in the attributes they contain and that the deviation in intents reflects the geometric distance between concepts in the data space, we thus define the intent-based distinction degree between granular concepts as follows.

**Definition 8.** Let  $D \in \mathbb{R}^{n \times d}$  denote the underlying discretized dataset from which the formal context  $\mathcal{K}$  is derived, where  $n$  and  $d$ , respectively, represent the numbers of objects and features of  $D$ . Let  $K$  denote the discretization granularity. For each granular concept  $C = (X, B) \in \mathcal{L}_g$ , we associate a representative intent vector  $Int(C) \in \mathbb{N}^d$ . For each feature  $a_i, 1 \leq i \leq d$ ,  $Int(C)(i) = j$  if all objects in  $X$  share the same discretized value  $j \in \{1, \dots, K\}$  on feature  $a_i$  and  $Int(C)(i) = 0$  otherwise.

For any two concepts  $C_1 = (X_1, B_1), C_2 = (X_2, B_2) \in \mathcal{L}_g$ , let  $T_{C_1, C_2} = \{t \mid Int(C_1)(t) \neq 0 \wedge Int(C_2)(t) \neq 0\}$  be the set of feature indices where both concepts have valid values. Denote  $\Phi(C_1, C_2) = \max_{t \in T_{C_1, C_2}} |Int(C_1)(t) - Int(C_2)(t)|$  if  $T_{C_1, C_2} \neq \emptyset$ . The intent-based distinction degree  $N$  is defined as

$$N(C_1, C_2) = \begin{cases} \Phi(C_1, C_2), & T_{C_1, C_2} \neq \emptyset \wedge \Phi(C_1, C_2) > 0, \\ k/2, & T_{C_1, C_2} \neq \emptyset \wedge \Phi(C_1, C_2) = 0, \\ k, & T_{C_1, C_2} = \emptyset. \end{cases} \quad (9)$$

**Remark 1.** The measure  $N$  in Formula (9) quantifies the explicit semantic conflict between granular concepts. In most cases, the intents of two granular concepts share common features ( $T_{C_1, C_2} \neq \emptyset$ ) and the application of the infinity norm emphasizes the most significant feature-level discrepancy between concepts. When two granular concepts have no shared features ( $T_{C_1, C_2} = \emptyset$ ), it indicates a substantial discrepancy between them. In addition, there also exist rare scenarios where different granular concepts share only a small number of attributes but have no overlap in most features and the average distance is adopted in such cases.

The initial set of granular concepts often contains a large number of concepts, many of which contribute marginally to characterizing dominant data structures. To focus on structurally representative patterns, we introduce a core concept extraction mechanism based on concept extent size.

**Definition 9.** Let  $\mathcal{K} = (U, A, I)$  be a formal context. For any formal concept  $(X, B) \in \mathcal{L}$ , we define the support of  $(X, B)$  as  $|X|/|U|$ . The granular concepts in  $\mathcal{L}_g$  are sorted in descending order of their support to generate the granular concept sequence  $(X_1, B_1), (X_2, B_2), \dots, (X_{|\mathcal{L}_g|}, B_{|\mathcal{L}_g|})$ . Given a proportion parameter  $\lambda \in (0, 1)$ , the set of core concepts is defined as:

$$\mathcal{L}_c = \{(X_i, B_i) \in \mathcal{L}_g \mid |X_i| \geq |X_{\lfloor \lambda |\mathcal{L}_g|}|\}. \quad (10)$$

**Remark 2.** The core concepts selected in Definition 9 capture dominant object-attribute patterns shared by a majority of samples. Concepts with small support tend to correspond to incidental structures. Focusing on core concepts can not only enhance the robustness of the model, but also significantly reduce the complexity of subsequent distance matrix calculation and outlier scoring. The screening of anomalous concepts mainly relies on the comparison of multi-dimensional differences between the granular concepts and the core concepts. The parameter  $\lambda$  identifies a concept in the granular concept sequence as the segmentation position and all granular concepts whose supports are greater than or equal to this one are selected as core concepts. This implies that the proportion of core concepts in the granular concepts will be slightly higher than  $\lambda$ , a phenomenon that occurs more frequently in categorical datasets. This observation, along with the sensitivity of core concept selection governed by  $\lambda$ , is analyzed in Section 5. Since  $|\mathcal{L}_c| \ll |\mathcal{L}_g|$  in most cases, this leads to a substantial reduction in runtime during the modeling phase.

#### 4.2. Outlier scoring

**Definition 10.** Let  $\mathcal{K} = (U, A, I)$  be a formal context and  $\{C_i = (X_i, B_i) \mid 1 \leq i \leq |\mathcal{L}_c|\}$  denote the set of core concepts. For any granular concept

#### Alg. 1 Structural Deviation Measurement

**Input:** Formal context  $\mathcal{K} = (U, A, I)$ , selection ratio  $\lambda \in (0, 1)$ , dataset  $D$

**Output:** Core concepts  $\mathcal{L}_c$ , discrepancy metrics  $M, N$

```

1: Generate granular concept space  $\mathcal{L}_g = \mathcal{L}_{og} \cup \mathcal{L}_{ag}$ ;
2: Calculate  $\{|X| \mid (X, B) \in \mathcal{L}_g\}$  and sort in descending order;
3: Select core subset  $\mathcal{L}_c$  satisfying the condition in Eq. (10);
4: for each pair  $(x, y) \in U \times U$  do
5:   Compute  $GCD(x, y)$  via Eq. (7);
6: end for
7: for each pair  $(C_i, C_j) \in \mathcal{L}_c \times \mathcal{L}_c$  do
8:   Compute extent-based structural discrepancy  $M_{ij}$  with Eq. (8);
9:   Compute intent-based semantic discrepancy  $N_{ij}$  with Eq. (9);
10: end for
11: return  $\mathcal{L}_c, M, N$ .
```

$C = (X, B) \in \mathcal{L}_g$ , the granular outlier degree ( $GOD$ ) of  $C$  is defined as:

$$GOD(X, B) = \frac{1}{|\mathcal{L}_c|} \sum_{i=1}^{|\mathcal{L}_c|} M(C, C_i) \cdot N(C, C_i), \quad (11)$$

where  $M$  and  $N$  calculate the extent-based and intent-based distinction degree between concepts via Formula (8) and Formula (9), respectively.

The multiplicative form in Eq. (11) ensures that a granular concept is regarded as highly deviant only when it exhibits significant discrepancies from most core concepts in both extent coverage and intent semantics.

**Definition 11.** Assume  $\mathcal{K} = (U, A, I)$  is a formal context. For any object  $x \in U$ , let  $\mathcal{L}_g(x) = \{(X, B) \in \mathcal{L}_g \mid x \in X\}$ . The core concept-based outlier factor ( $CCOF$ ) of  $x$  is defined as:

$$CCOF(x) = \frac{1}{|\mathcal{L}_g(x)|} \sum_{(X_j, B_j) \in \mathcal{L}_g(x)} GOD(X_j, B_j) \cdot W(X_j, B_j), \quad (12)$$

where the weight  $W(X_j, B_j)$  is given by  $W(X_j, B_j) = 1 - (|X_j|/|U|)^{\frac{1}{3}}$ .

Core concept-based analysis primarily captures structural deviation arising from object participation in concept extents. However, such structural measures may overlook fine-grained deviations that originate from statistically rare attribute configurations. In particular, objects associated with globally infrequent attributes may remain embedded within dominant granular concepts and thus receive low  $CCOF$ . To address this limitation, we introduce a measure, termed global probability deviation factor ( $GPDF$ ), to quantify attribute-level statistical rarity from an information-theoretic perspective.

**Definition 12.** Let  $\mathcal{K} = (U, A, I)$  be a formal context. For each attribute  $a \in A$ , its global occurrence probability is defined as:

$$P(a) = \frac{|\{x \in U \mid (x, a) \in I\}|}{|U|}. \quad (13)$$

For each object  $x \in U$ , the global probability deviation factor of  $x$  is computed as:

$$GPDF(x) = \sum_{a \in A} \mathbb{I}((x, a) \in I) \log \left( \frac{1}{P(a) + \varepsilon} \right), \quad (14)$$

where  $\mathbb{I}(\cdot)$  denotes the indicator function and  $\varepsilon$  is a small constant ensuring numerical stability.

In the above definition, the term  $\log(1/P(a))$  corresponds to the self-information of observing attribute  $a$ . This formulation can be viewed as a standard additive approximation of joint rarity and assigns higher scores to objects possessing theoretically rare feature combinations. To jointly exploit structural isolation and attribute-level statistical rarity, we combine the core concept-based outlier factor with the global probability deviation factor.



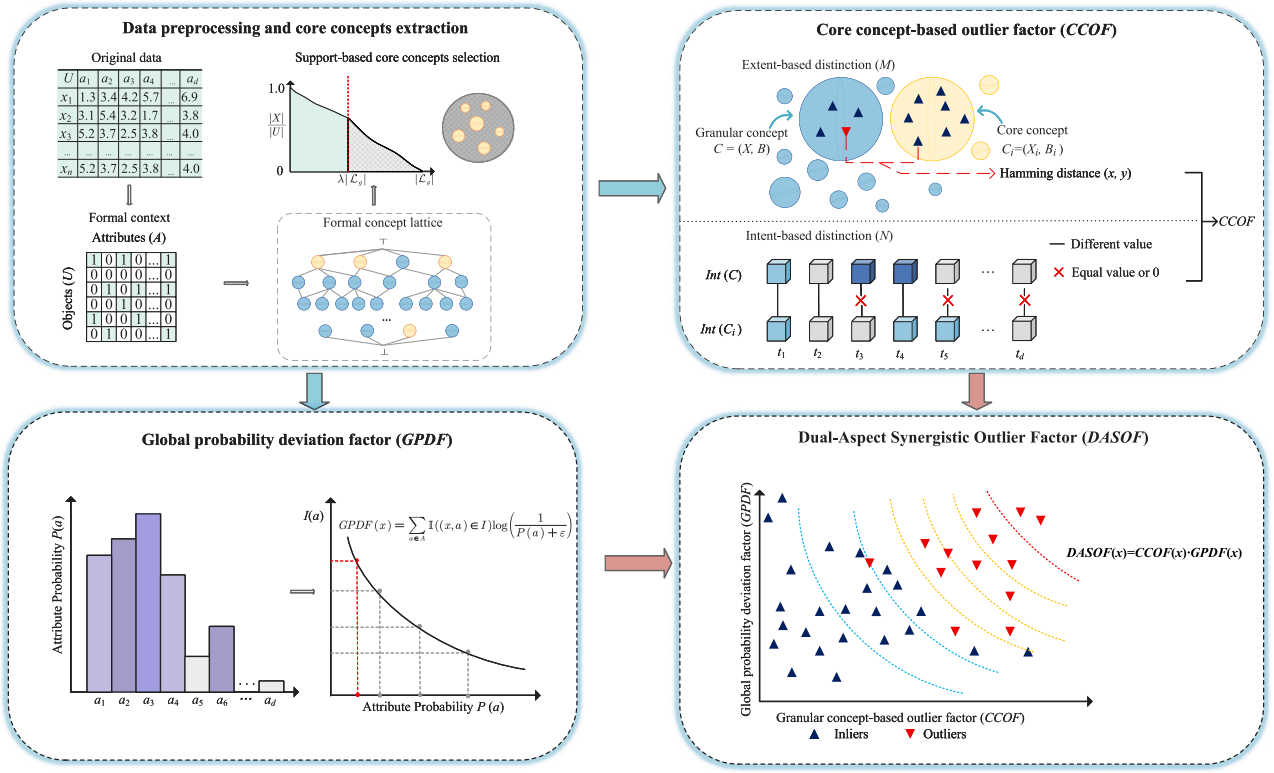


Fig. 1. Overall framework of DASOD.

**Alg. 2** Dual-Aspect Synergistic Outlier Factor Calculating**Input:** Dataset  $D$ , core concepts  $\mathcal{L}_c$ , discrepancy metrics  $M, N$ **Output:** Outlier score vector  $DASOF$ 

```

1: for each  $(X, B) \in \mathcal{L}_g$  do
2:   Compute  $GOD(X, B)$  by aggregating  $M$  and  $N$  via Eq. (11);
3:   Compute weight  $W(X, B)$  based on extent size;
4: end for
5: for each object  $x \in U$  do
6:   Compute  $CCOF(x)$  by Eq. (12);
7:   Compute  $GPDF(x)$  using self-information summation via Eq. (14);
8: end for
9: for each  $x \in U$  do
10:  Compute  $DASOF(x)$  by Eq. (15);
11: end for
12: return  $DASOF$ .
```

**Definition 13** (*Dual-Aspect Synergistic Outlier Factor*). Let  $D$  be a dataset from which the formal context  $\mathcal{K} = (U, A, I)$  is derived. The dual-aspect synergistic outlier factor of arbitrary  $x \in U$  is defined as:

$$DASOF(x) = CCOF(x) \cdot GPDF(x). \quad (15)$$

**Remark 3.** The multiplicative formulation in (15) implicitly enforces a joint constraint on the dual-aspect of structural and statistical deviations. Objects exhibiting high structural isolation but low attribute-level rarity, or vice versa, are assigned moderate outlier scores. As a result, false positives caused by one-sided deviations can be effectively suppressed.

Compared with the linear superposition widely adopted in existing literature, this approach eliminates unnecessary weighting factors, thereby greatly reducing the computational complexity. Meanwhile, it achieves nearly equivalent or slightly inferior experimental

results on various datasets. This advantage is attributed to the dualization processing of objects and attributes in FCA. Detailed comparative experiments are presented in Section 5.4.

To enable the algorithm to adapt easily to different types of datasets, we perform the same  $K$ -width equal discretization for categorical features in categorical datasets and mixed datasets as that for numerical features when constructing the data-induced formal context. In addition, rows and columns with all elements being 0 or 1 are removed during the formal context generation, which ensures that the formal context processed by the algorithm is regular. For categorical features, this operation can be regarded as a new grouping of all samples, where samples with the same category are always assigned to the same discretized interval.

**4.3. Time complexity analysis**

The computational complexity of the DASOD framework is determined by four sequential phases (see Fig. 1). Let  $|U|$ ,  $d$ ,  $|A|$ ,  $|\mathcal{L}_g|$ , and  $|\mathcal{L}_c|$  denote the number of objects, original features, discretized attributes, granular concepts, and core concepts, respectively. During the initial preprocessing phase, formal context construction and the generation of granular concepts require  $O(|U|^2|A| + |U||A|^2)$ . Subsequently, the core concept extraction process involving support calculation and descending sorting takes  $O(|U||\mathcal{L}_g| + |\mathcal{L}_g|\log|\mathcal{L}_g|)$ . In the structural modeling phase, the computation of  $M$  is optimized via matrix associativity to  $O(|U|^2|\mathcal{L}_g||\mathcal{L}_c|)$ , while the derivation of  $N$  results in  $O(|\mathcal{L}_g||\mathcal{L}_c|d)$ . Finally, the scoring stage which includes  $GOD$ ,  $CCOF$ , and  $GPDF$  computation requires  $O(|\mathcal{L}_g||\mathcal{L}_c| + |U||\mathcal{L}_g| + |U||A|)$ .

By combining these phases and omitting lower-order terms, the overall time complexity is bounded by  $O(|U|^2|A| + |U||A|^2 + |U|^2|\mathcal{L}_g||\mathcal{L}_c| + |\mathcal{L}_g||\mathcal{L}_c|d)$ . In practical applications, we have  $|\mathcal{L}_g| \ll |\mathcal{L}_c| \leq |U| + |A|$ . Consequently, the dominant cost remains strictly polynomial, which ensures the scalability of the framework for large-scale tasks while avoiding the exponential state-space explosion typical of traditional methods.

**Table 1**  
Characteristics of the 16 benchmark datasets.

| NO. | Dataset    | Samples | Features | Outliers | Ratio (%) | Feature type |
|-----|------------|---------|----------|----------|-----------|--------------|
| 1   | adult      | 37,793  | 14       | 3779     | 10.00     | Mixed        |
| 2   | arrhythmia | 452     | 279      | 66       | 14.60     | Mixed        |
| 3   | audiology  | 226     | 69       | 57       | 25.22     | Mixed        |
| 4   | banknote   | 1372    | 4        | 610      | 44.46     | Numerical    |
| 5   | chess      | 1814    | 36       | 145      | 7.99      | Categorical  |
| 6   | credit     | 425     | 15       | 42       | 9.88      | Mixed        |
| 7   | diabetes   | 526     | 8        | 26       | 4.94      | Numerical    |
| 8   | german     | 714     | 20       | 14       | 1.96      | Mixed        |
| 9   | horse      | 256     | 27       | 12       | 4.69      | Mixed        |
| 10  | mushroom   | 4429    | 22       | 221      | 4.99      | Categorical  |
| 11  | nursery    | 12,960  | 8        | 330      | 2.55      | Categorical  |
| 12  | pendigits  | 6870    | 16       | 156      | 2.27      | Numerical    |
| 13  | pima       | 555     | 9        | 55       | 9.91      | Numerical    |
| 14  | shuttle    | 49,097  | 9        | 3511     | 7.15      | Numerical    |
| 15  | tic        | 695     | 9        | 69       | 9.93      | Categorical  |
| 16  | wholesale  | 440     | 7        | 142      | 32.27     | Numerical    |

**Table 2**  
Summary of baseline methods and their parameter settings.

| Method              | Type          | Parameter settings                     |
|---------------------|---------------|----------------------------------------|
| 3W-SGBD (2026) [37] | GrC           | $\lambda \in [0.1, 1.0]$ , step 0.1    |
| EDROD (2025) [29]   | Density       | $k \in [10, 100]$ , step 10            |
| GBDO (2025) [35]    | GrC           | $k \in [2, 31]$ , step 3               |
| MGBOD (2025) [36]   | GrC           | $\sigma \in [0.1, 1.0]$ , step 0.1     |
| TIEOD (2025) [21]   | GrC           | $b \in [2, 10]$ , step 1               |
| DIF (2023) [31]     | Deep learning | $n_{ensemble} \in \{10, 25, 50, 100\}$ |
| ECOD (2023) [24]    | Statistical   | Parameter-free                         |
| GCOD (2023) [20]    | GrC           | $b \in [2, 10]$ , step 1               |

## 5. Experiments

To systematically assess the effectiveness of the proposed outlier detection method, extensive experiments were conducted on 16 benchmark datasets. Under unified evaluation protocols, the proposed approach is compared with eight state-of-the-art baseline methods. Comprehensive analyses were performed from multiple perspectives, including overall detection performance, robustness with respect to key parameter variations, comparisons with the linear superposition method, and representative case studies. The results consistently demonstrate the competitiveness and practical applicability of the proposed method.

### 5.1. Experimental settings

Experiments were performed on various publicly available benchmark datasets. The selected datasets exhibit substantial diversity in terms of sample size, feature dimensionality, outlier ratio and data characteristics, including continuous-valued features, mixed-type attributes and categorical attributes. This diversity enables a comprehensive evaluation of the proposed method under different outlier detection scenarios. The fundamental statistics of each dataset, including the number of samples, feature dimensionality and outlier ratio, are summarized in Table 1.

To comprehensively evaluate the proposed DASOD, we compare it with eight representative outlier detection baselines covering deep representation learning, statistical modeling, density-based approaches, and granular computing. To ensure a rigorous and fair evaluation, all baselines were evaluated using publicly available open-source implementations or were carefully reproduced according to the descriptions provided in their original papers. Stochastic algorithms were averaged over ten runs with different random seeds. Extensive parameter searches were conducted within the ranges recommended in the original literature and the corresponding search spaces are summarized in Table 2.

We adopt widely used evaluation metrics in the literature to assess the detection performance of competing algorithms, including the ROC

curve, Area Under Curve (AUC) [20], and Max F1-score [46]. An ROC curve that lies closer to the top-left corner of the coordinate axis, along with higher AUC and Max F1-score values, indicates superior algorithm performance. These metrics capture performance across all possible thresholds.

### 5.2. Experimental results and discussion

A systematic comparison was conducted against eight representative state-of-the-art outlier detection methods across 16 publicly available benchmark datasets. We first plotted the ROC curves for comparison in Fig. 2. On most datasets, the ROC curve of our method lies closer to the top-left corner of the plot, indicating its superior detection performance. The quantitative results of AUC and MaxF1 score are respectively presented in Tables 3 and 4, where the bold labels optimal values on datasets.

In terms of AUC, which evaluates the quality of outlier ranking, DASOD achieves the highest average score of 0.9093 among all compared methods. This performance consistently surpasses competing approaches, including TIEOD and GCOD, which obtain average AUC values of 0.8854 and 0.8749, respectively. Notably, DASOD attains the best AUC performance on 15 out of the 16 datasets, demonstrating an exceptionally strong and stable ranking capability across diverse data distributions.

With respect to detection accuracy under an optimal decision threshold as measured by the MaxF1 score, DASOD achieves the highest average score of 0.6480. This result exceeds the second-best method, TIEOD, by a significant margin of approximately 5 percent. Such results indicate that DASOD generates more discriminative outlier scores and facilitates a clearer separation between normal and anomalous samples. Furthermore, DASOD exhibits robust performance even on challenging datasets where traditional methods struggle, further confirming its practical efficacy.

To evaluate the statistical significance of the observed performance improvements, we conducted a Friedman test followed by a Nemenyi post-hoc analysis at a significance level of 0.05. The Critical Difference diagrams are illustrated in Fig. 3. In both AUC and MaxF1 evaluations, DASOD achieves the lowest average rank, specifically 1.0625 for AUC and 1.2188 for MaxF1, among all compared methods. The rank differences between DASOD and most baseline approaches exceed the critical difference threshold, indicating statistically significant superiority. These results provide strong evidence that DASOD maintains consistently strong performance across datasets with varying attribute types and complexity.

### 5.3. Parameter sensitivity analysis

The stability and efficiency of DASOD are evaluated through an analysis of the joint impact of discretization granularity  $K$  and the core concept selection ratio  $\lambda$ . Within this framework, the parameter  $K$  determines the refinement of the feature space, whereas  $\lambda$  serves as a hyper-parameter introduced by the DASOD algorithm to control the selection of dominant granular structures. In our experimental setup, the value of  $\lambda$  varies from 1% to 10% and is compared against the baseline where  $\lambda$  equals 100% to represent the utilization of the full concept set. Furthermore, the discretization parameter  $K$  is selected from the set of values 2, 4, 6, 8, and 10 to provide a comprehensive assessment of model performance.

The experimental results across the sixteen benchmark datasets indicate that the AUC performance exhibits a clear trend of rapid convergence as  $\lambda$  increases. By prioritizing these core patterns and filtering out a large number of redundant granular structures, DASOD maintains high detection accuracy while significantly reducing computational complexity. As a robust algorithm parameter,  $\lambda$  exhibits high stability. Therefore, we fix  $\lambda$  at 5% in all subsequent experiments to maintain

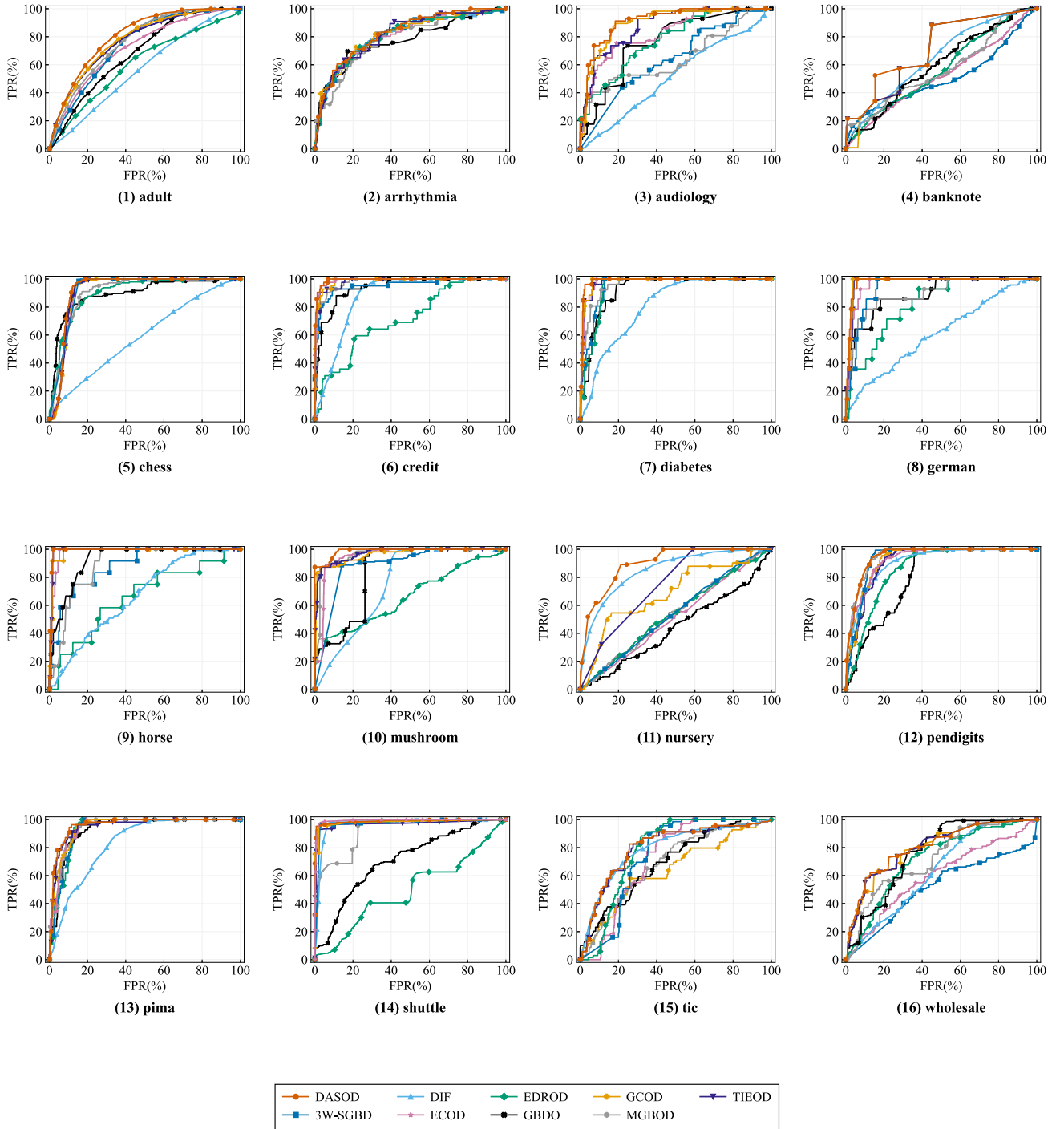


Fig. 2. Experimental comparison results of ROC curves for 9 methods on 16 datasets.

model simplicity and ensure consistent performance across all domains. In accordance with Remark 2, the cardinality of the selected core concepts typically surpasses the theoretical threshold. The following empirical results on partial datasets substantiate this observation (see Table 5).

The parameter  $K$  represents the discretization granularity. As illustrated in Fig. 4, the optimal  $K$  varies significantly across different datasets, depending on sample size, feature dimensionality, feature type, and inherent data distribution. In practice, the determination of the optimal  $K$  is fundamentally a trade-off between the demand for information resolution and the constraints imposed by spatial sparsity.

This paper investigates the configuration of the parameter  $K$  and reduces its search scope.

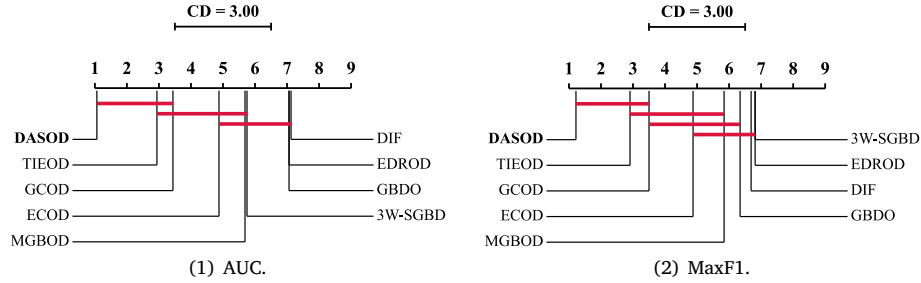
Firstly, the optimal number of  $K$  for equal-width discretization depends on the feature type of the dataset. Secondly, the sample-to-feature ratio  $R = |U|/d$  is used to evaluate the sample density in multi-dimensional space. Thirdly, considering that numerous samples in the dataset share identical values on a single feature, we introduce the average number of distinct values across all feature columns as a reference for parameter determination. Let  $H = 1/d \sum_{i=1}^d |V_i|$ ,  $1 \leq i \leq d$ , where  $V_i$  represents the set of unique values within the  $i$ -th feature column. Based on theoretical analysis and empirical investigation, we

**Table 3**  
AUC comparison across 16 benchmark datasets.

| Dataset        | 3W-SGBD       | EDROD         | MGBOD         | GBDO          | TIEOD         | DIF           | ECOD          | GCOD          | DASOD         |
|----------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| adult          | 0.7430        | 0.6078        | 0.7636        | 0.6856        | 0.7701        | 0.5690        | 0.7222        | 0.7767        | <b>0.8056</b> |
| arrhythmia     | 0.8068        | 0.8100        | 0.7865        | 0.7749        | 0.8140        | 0.8067        | 0.8071        | 0.8088        | <b>0.8249</b> |
| audiology      | 0.6420        | 0.7792        | 0.6503        | 0.7693        | 0.8781        | 0.5090        | 0.8326        | 0.9012        | <b>0.9167</b> |
| banknote       | 0.5023        | 0.5693        | 0.5727        | 0.5960        | 0.7043        | 0.6498        | 0.5220        | 0.6913        | <b>0.7277</b> |
| chess          | 0.9221        | 0.9001        | 0.9050        | 0.9028        | 0.9185        | 0.5902        | 0.9150        | 0.9169        | <b>0.9234</b> |
| credit         | 0.9547        | 0.7142        | 0.9769        | 0.9386        | 0.9815        | 0.8730        | 0.9903        | 0.9849        | <b>0.9929</b> |
| diabetes       | 0.9497        | 0.9405        | 0.9590        | 0.9255        | 0.9802        | 0.8073        | 0.9787        | 0.9828        | <b>0.9881</b> |
| german         | 0.9363        | 0.8237        | 0.8914        | 0.8872        | 0.9773        | 0.6096        | 0.9655        | 0.9799        | <b>0.9805</b> |
| horse          | 0.8798        | 0.6578        | 0.8893        | 0.9254        | 0.9901        | 0.6663        | 0.9805        | 0.9853        | <b>0.9904</b> |
| mushroom       | 0.8876        | 0.6615        | 0.9619        | 0.8345        | 0.9701        | 0.7525        | 0.9489        | 0.9634        | <b>0.9882</b> |
| nursery        | 0.5415        | 0.5393        | 0.5377        | 0.4653        | 0.7442        | 0.8503        | 0.5153        | 0.7105        | <b>0.8991</b> |
| pendigits      | 0.9320        | 0.8564        | 0.9488        | 0.8035        | 0.9126        | 0.9068        | 0.9274        | 0.9225        | <b>0.9494</b> |
| pima           | 0.9352        | 0.9339        | 0.9438        | 0.9315        | 0.9513        | 0.8270        | 0.9472        | 0.9496        | <b>0.9610</b> |
| shuttle        | 0.9733        | 0.4965        | 0.9190        | 0.7096        | 0.9720        | 0.9682        | <b>0.9929</b> | 0.9820        | 0.9890        |
| tic            | 0.7474        | 0.7839        | 0.6945        | 0.7065        | 0.7999        | 0.7929        | 0.7235        | 0.6453        | <b>0.8051</b> |
| wholesale      | 0.5156        | 0.7191        | 0.7231        | 0.7622        | 0.8020        | 0.6367        | 0.5989        | 0.7970        | <b>0.8063</b> |
| <b>Average</b> | <b>0.8043</b> | <b>0.7371</b> | <b>0.8202</b> | <b>0.7886</b> | <b>0.8854</b> | <b>0.7385</b> | <b>0.8355</b> | <b>0.8749</b> | <b>0.9093</b> |

**Table 4**  
MaxF1 comparison across 16 benchmark datasets.

| Dataset        | 3W-SGBD       | EDROD         | MGBOD         | GBDO          | TIEOD         | DIF           | ECOD          | GCOD          | DASOD         |
|----------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| adult          | 0.2954        | 0.2303        | 0.3143        | 0.2539        | 0.3386        | 0.2024        | 0.2986        | 0.3422        | <b>0.3625</b> |
| arrhythmia     | 0.5000        | 0.5098        | 0.4541        | 0.5169        | 0.5135        | 0.5201        | 0.5029        | 0.5091        | <b>0.5324</b> |
| audiology      | 0.4623        | 0.5479        | 0.5091        | 0.5970        | 0.6916        | 0.4123        | 0.6515        | 0.7353        | <b>0.7568</b> |
| banknote       | 0.6199        | 0.6355        | 0.6414        | 0.6350        | <b>0.7225</b> | 0.6469        | 0.6225        | <b>0.7225</b> | <b>0.7225</b> |
| chess          | 0.5257        | 0.4971        | 0.4796        | 0.5430        | 0.5444        | 0.1815        | 0.5422        | 0.5495        | <b>0.5673</b> |
| credit         | 0.7640        | 0.3467        | 0.8250        | 0.6905        | 0.8444        | 0.4669        | 0.8780        | 0.8222        | <b>0.8941</b> |
| diabetes       | 0.4773        | 0.4587        | 0.5714        | 0.4444        | 0.7586        | 0.2659        | 0.7213        | 0.7500        | <b>0.8070</b> |
| german         | 0.2727        | 0.2500        | 0.3043        | 0.3214        | 0.4828        | 0.1302        | 0.4286        | 0.5185        | <b>0.5217</b> |
| horse          | 0.4242        | 0.2069        | 0.3529        | 0.4545        | <b>0.8276</b> | 0.1880        | 0.6667        | 0.8148        | <b>0.8276</b> |
| mushroom       | 0.3874        | 0.3456        | 0.7338        | 0.3456        | 0.7562        | 0.1983        | 0.6101        | 0.8176        | <b>0.9324</b> |
| nursery        | 0.0547        | 0.0581        | 0.0567        | 0.0509        | 0.1735        | 0.2650        | 0.0521        | 0.0896        | <b>0.3650</b> |
| pendigits      | 0.2509        | 0.1565        | <b>0.3737</b> | 0.1291        | 0.2612        | 0.2956        | 0.3669        | 0.2671        | 0.3533        |
| pima           | 0.5917        | 0.6082        | 0.6241        | 0.6241        | 0.7018        | 0.3913        | 0.6400        | 0.6349        | <b>0.7227</b> |
| shuttle        | 0.8313        | 0.1598        | 0.6048        | 0.2510        | 0.9141        | 0.7388        | 0.8795        | 0.8622        | <b>0.9282</b> |
| tic            | 0.3473        | 0.3708        | 0.2731        | 0.3066        | 0.4018        | 0.4081        | 0.3105        | 0.2930        | <b>0.4103</b> |
| wholesale      | 0.4880        | 0.6015        | 0.5916        | 0.6419        | 0.6484        | 0.5636        | 0.4978        | 0.6435        | <b>0.6645</b> |
| <b>Average</b> | <b>0.4558</b> | <b>0.3740</b> | <b>0.4819</b> | <b>0.4254</b> | <b>0.5988</b> | <b>0.3672</b> | <b>0.5418</b> | <b>0.5858</b> | <b>0.6480</b> |



**Fig. 3.** CD plot for all compared algorithms.

**Table 5**  
Comparison of selected core concept quantity and theoretical threshold.

| Dataset    | $K$ | $ \mathcal{L}_g $ | $ \mathcal{L}_c $ | $5\% \mathcal{L}_g $ | $ X_{\{ \mathcal{L}_g \}} $ | Feature type |
|------------|-----|-------------------|-------------------|----------------------|-----------------------------|--------------|
| adult      | 7   | 20936             | 1248              | 1047                 | 5                           | Mixed        |
| arrhythmia | 7   | 1543              | 78                | 77                   | 444                         | Mixed        |
| audiology  | 4   | 315               | 21                | 16                   | 225                         | Mixed        |
| mushroom   | 9   | 4121              | 508               | 206                  | 2                           | Categorical  |
| nursery    | 2   | 272               | 16                | 14                   | 4320                        | Categorical  |
| pima       | 7   | 516               | 27                | 26                   | 35                          | Numerical    |

propose the following principles for setting  $K$ .

$$K \in \begin{cases} [2, 4], & \text{if } (D = \text{Categorical}) \vee (R < 10 \wedge H \leq 5), \\ [5, 8], & \text{if } D = \text{Mixed}, \\ [7, 10], & \text{if } (D = \text{Numerical}) \vee (R \geq 10 \wedge H > 5). \end{cases} \quad (16)$$

#### 5.4. Comparison with the linear superposition strategy

To validate the balance between detection performance and computational efficiency, we compared DASOD with two exhaustive search schemes based on linear superposition, i.e., using  $\alpha CCOF + (1-\alpha)GPDP$  as the final outlier score. Strategy A involves a weight grid search with a fixed  $K$  value, while Strategy B conducts a comprehensive search over the dual  $K$  value affecting  $CCOF$  and  $GPDP$ , together with the weight  $\alpha$ . According to the results in Table 6, Strategy B achieves the highest numerical performance, yet the average improvements in AUC and MaxF1 over DASOD are limited to 0.0080 and 0.0100, respectively. These marginal gains come at the cost of a search space requiring 891 iterations, which represents a computational burden nearly 90 times



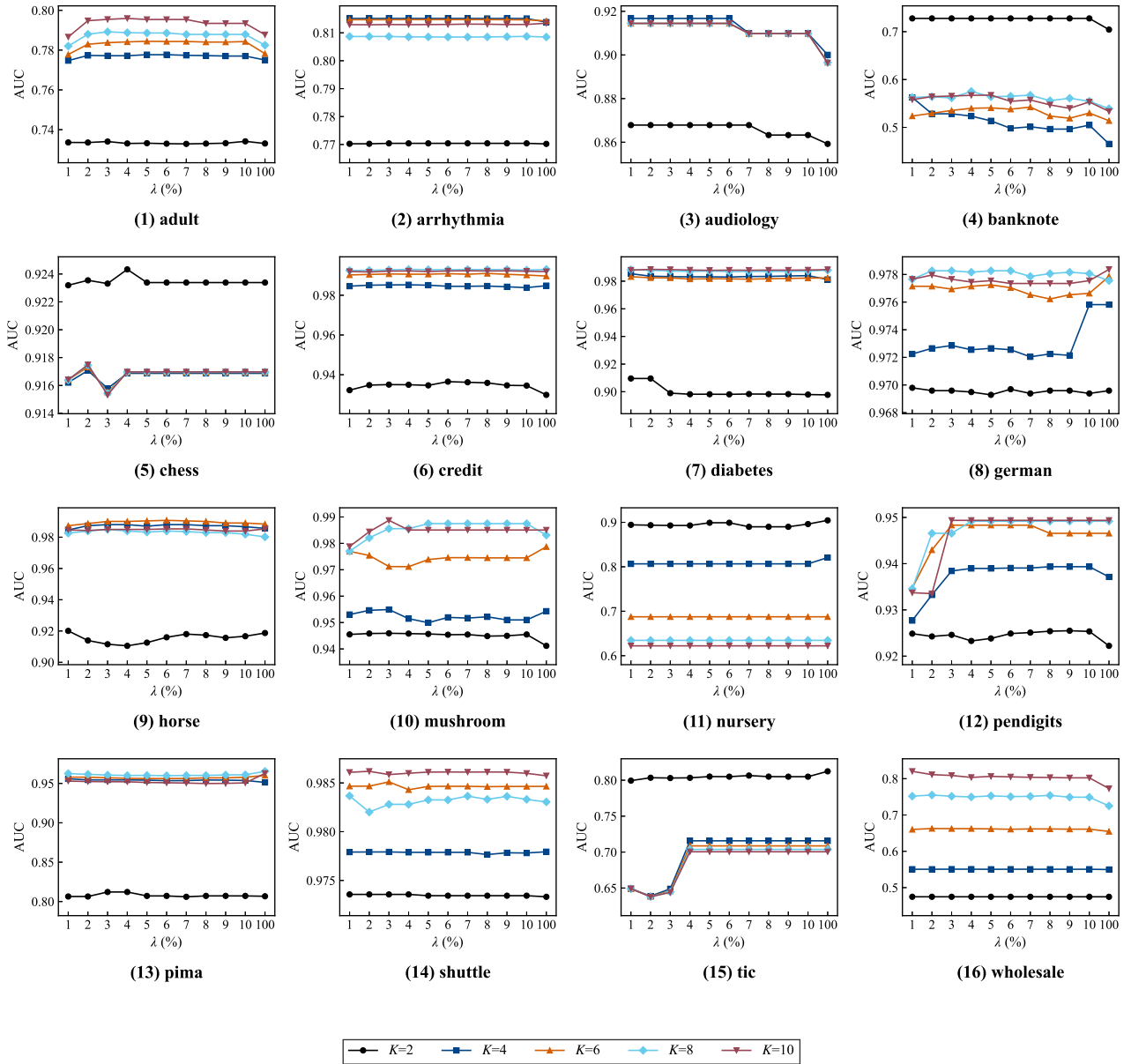


Fig. 4. AUC results of the proposed DASOD under various parameter configurations on 16 benchmark datasets.

greater than DASOD. Since unsupervised tasks lack ground-truth labels for parameter tuning in real-world deployment, the ability of DASOD to achieve over 99% of the optimal performance without manual intervention demonstrates superior robustness. This supports our selection of fusion strategy instead of resource-intensive search methods.

### 5.5. Case study

To evaluate the interpretability of DASOD, we conducted a case study on the Wisconsin breast cancer dataset [47]. Since numerical outlier scores alone are insufficient for clinical adoption, DASOD addresses this black-box limitation by aligning hybrid outlier evidence with human-interpretable semantics. We demonstrate this via a three-level framework: (1) structural distribution analysis for granular concepts, (2) semantic alignment with cytological criteria, and (3) multi-perspective decomposition of hybrid evidence.

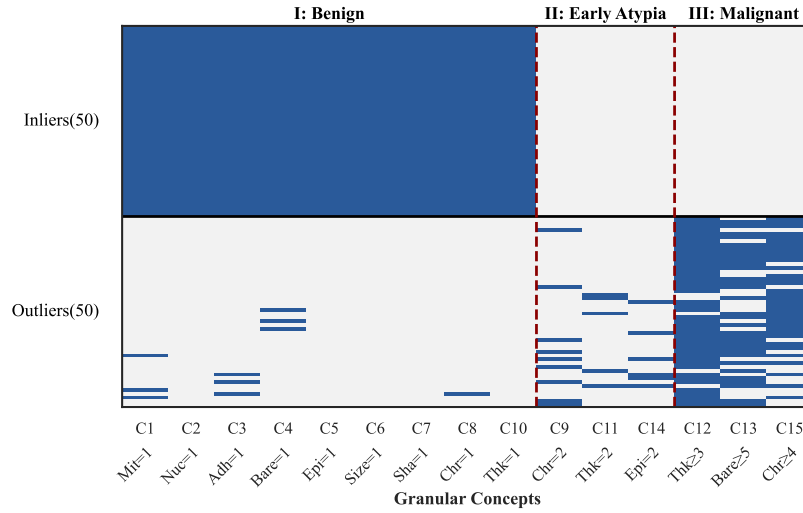
Fig. 5 shows the coverage of the 50 lowest-scoring inliers and 50 highest-scoring outliers over core concepts. A clear block-diagonal structure indicates a strong separation between benign and malignant

states. Group II (early atypia) covers no low-scoring inliers, suggesting that typical benign samples are strictly confined to Group I, where even slight deviations signal early pathological changes. In contrast, high-scoring outliers display fragmented coverage across Groups II and III, reflecting a progressive pathological transition. This pattern is consistent with the cytological principle of pleomorphism, where malignancy emerges as a disordered deviation from normal cellular uniformity [48].

Table 7 validates the semantic consistency of the identified granular structures by aligning the core rules extracted by DASOD with established cytological grading criteria [47]. Rules in Group I have the largest extents, representing canonical patterns of cellular stability, and consistently cover samples with the lowest outlier scores. By contrast, Group III captures high-risk patterns. For instance, C13 defines a Bare Nuclei threshold of 5 and C15 specifies a Chromatin level of 4, both of which are classic malignant biomarkers. DASOD assigns high outlier scores to samples satisfying these threshold constraints. This confirms that its reasoning follows domain-specific semantics, rather than relying on spurious numerical correlation. The gradual transition

**Table 6**  
Comparison with the linear superposition strategy based on AUC and MaxF1.

| Dataset        | AUC           |               |               | MaxF1         |               |               |
|----------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                | DASOD         | Strategy A    | Strategy B    | DASOD         | Strategy A    | Strategy B    |
| adult          | 0.8056        | 0.8215        | 0.8215        | 0.3625        | 0.3710        | 0.3710        |
| arrhythmia     | 0.8249        | 0.8272        | 0.8273        | 0.5324        | 0.5294        | 0.5294        |
| audiology      | 0.9167        | 0.9234        | 0.9234        | 0.7568        | 0.7706        | 0.7706        |
| banknote       | 0.7277        | 0.7277        | 0.7277        | 0.7225        | 0.7225        | 0.7225        |
| chess          | 0.9234        | 0.9235        | 0.9235        | 0.5673        | 0.5691        | 0.5691        |
| credit         | 0.9929        | 0.9932        | 0.9939        | 0.8941        | 0.8941        | 0.9195        |
| diabetes       | 0.9881        | 0.9893        | 0.9902        | 0.8070        | 0.8197        | 0.7879        |
| german         | 0.9805        | 0.9811        | 0.9816        | 0.5217        | 0.5128        | 0.5116        |
| horse          | 0.9904        | 0.9908        | 0.9918        | 0.8276        | 0.8462        | 0.8462        |
| mushroom       | 0.9882        | 0.9891        | 0.9891        | 0.9324        | 0.9061        | 0.9061        |
| nursery        | 0.8991        | 0.9017        | 0.9343        | 0.3650        | 0.3650        | 0.3918        |
| pendigits      | 0.9494        | 0.9532        | 0.9534        | 0.3533        | 0.3342        | 0.3351        |
| pima           | 0.9610        | 0.9613        | 0.9655        | 0.7227        | 0.7227        | 0.7656        |
| shuttle        | 0.9890        | 0.9896        | 0.9910        | 0.9282        | 0.9408        | 0.9734        |
| tic            | 0.8051        | 0.8094        | 0.8398        | 0.4103        | 0.4231        | 0.4516        |
| wholesale      | 0.8063        | 0.8223        | 0.8223        | 0.6645        | 0.6767        | 0.6767        |
| <b>Average</b> | <b>0.9093</b> | <b>0.9128</b> | <b>0.9173</b> | <b>0.6480</b> | <b>0.6502</b> | <b>0.6580</b> |



**Fig. 5.** Structural perspective of the Wisconsin breast cancer dataset.

from Group I to Group III shows that the FCA-based mechanism naturally preserves the monotonic trend between structural deviation and malignancy level.

Fig. 6(1) provides a micro-level explanation by comparing the intent vectors of representative samples against the Core Prototype which serves as the gray baseline for the ideal benign state. While the inliers represented by the blue line remain highly congruent with the prototype, the outliers shown in red exhibit coordinated deviations across multiple morphological features, with certain feature combinations showing pronounced synergistic anomalies. This systemic manifestation of malignant transformation can be effectively captured by *GPDF*. Although clinical diagnosis is often a holistic synthesis of multiple indicators, the adoption of the infinity norm  $L_\infty$  in the deviation measure  $N(\cdot, \cdot)$  is strategically chosen to act as a critical trigger mechanism. This design ensures that the most extreme feature-specific breach is preserved rather than being diluted by the averaging effects of other distance metrics. Such a focus on maximal deviation aligns with the conservative screening principle in pathology where a single highly atypical morphological feature is sufficient to justify intensive clinical investigation regardless of the states of other attributes [49].

The joint distribution of *CCOF* and *GPDF* is illustrated in Fig. 6(2). This scatter plot reveals a positive linear correlation between the two indicators, indicating that significant malignancies typically trigger both local structural conflicts and global distribution rarity simultaneously. Despite this correlation, the fusion strategy adopted in

Eq. (15) remains essential. It acts as a dual validation mechanism to effectively suppress potential false positives caused by the deviation of a single indicator. True outliers are mainly concentrated in the region where both types of deviations are relatively high.

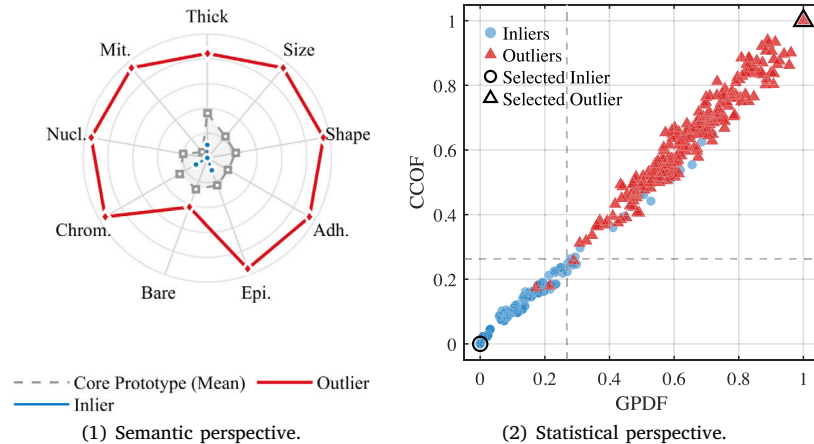
## 6. Conclusion

This paper has proposed a novel GrC outlier detection framework DASOD based on FCA, which integrates the structural deviation of concept lattices and the statistical rarity of attributes. Extensive experiments on 16 benchmark datasets have shown that DASOD outperforms eight state-of-the-art baseline algorithms in terms of AUC and MaxF1. Crucially, detailed case studies have revealed that the granular rules mined by this method are highly consistent with domain-specific prior knowledge, providing multi-level interpretability that is often lacking in black-box models. With its superior detection performance and interpretability, the proposed framework can be effectively applied to real-world scenarios that require reliable and interpretable outlier detection.

Despite the above advantages, DASOD still exhibits some notable limitations. First, the optimal value of the discretization parameter  $K$  varies greatly across different datasets. Although this paper has investigated the configuration of parameter  $K$  and narrowed its search range, it cannot be explicitly incorporated into the outlier factor as

**Table 7**  
Semantic alignment between discovered granular rules and cytological standards.

| Granular concepts ID              | Extent | Granular rule | Alignment with cytological standards                                         |
|-----------------------------------|--------|---------------|------------------------------------------------------------------------------|
| I. Strict benign primitives       |        |               |                                                                              |
| C1                                | 598    | Mitoses = 1   | Represents the inert state with lowest mitotic activity.                     |
| C2                                | 468    | Nucl. = 1     | Indicates nuclear regularity with normal and inconspicuous nucleoli.         |
| C3                                | 451    | Adh. = 1      | Reflects structural cohesion with high marginal adhesion.                    |
| C4                                | 432    | Bare = 1      | Indicates cytoplasmic retention where nuclei are surrounded by cytoplasm.    |
| C5                                | 420    | Epi. = 1      | Shows epithelial stability with standard single epithelial cell size.        |
| C6                                | 418    | Size = 1      | High uniformity characterized by consistent cell size.                       |
| C7                                | 404    | Shape = 1     | High uniformity characterized by consistent cell shape.                      |
| C8                                | 310    | Chrom. = 1    | Fine texture with uniform chromatin distribution.                            |
| C10                               | 189    | Thick = 1     | Monolayer arrangement without clumping.                                      |
| II. Early pathological deviations |        |               |                                                                              |
| C9                                | 200    | Chrom. = 2    | Texture bridging showing slight coarsening remaining within benign limits.   |
| C11                               | 183    | Thick = 2     | Mild multi-layering distinct from malignant clumps.                          |
| C14                               | 119    | Epi. = 2      | Slight enlargement showing minor epithelial size variation within tolerance. |
| III. Malignant indicators         |        |               |                                                                              |
| C12                               | 161    | Thick = 3     | Critical threshold boundary often breached by anomalous outliers.            |
| C13                               | 141    | Bare = 5      | Loss of cytoplasm (Bare Nuclei), a malignancy sign.                          |
| C15                               | 99     | Chrom. = 4    | Coarse texture indicating aggressive nuclear patterns.                       |



**Fig. 6.** Interpretability analysis of DASOD on the Wisconsin breast cancer dataset. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

an explicit variable. This makes it difficult to construct an adaptive parameter selection mechanism. Second, granular concept construction involves high time complexity, which restricts its scalability on large-scale datasets and high-dimensional sparse data scenarios. To tackle these limitations, future research will focus on developing an adaptive discretization mechanism. Appropriate clustering algorithms will be introduced to overcome the drawbacks of fine-grained representation, and further improve the applicability and computational efficiency for large-scale data.

Recently, Yan et al. [50] have provided a research paradigm for the integration of concept-cognitive learning and graph data processing. They regarded the extents of formal concepts as subgraph patterns, fused node structural information and attribute features in a hierarchical concept space, and further established a graph network semi-supervised concept-cognitive learning model suitable for node classification tasks. These explorations pave the way for extending concept-cognitive learning models to graph outlier detection, which will be studied in our future work.

#### CRedit authorship contribution statement

**Jinhai Li:** Writing – original draft, Methodology, Funding acquisition. **Jiayao Shi:** Software, Resources, Data curation. **Shen Li:** Writing – review & editing, Funding acquisition. **Yu Chen:** Writing – review & editing, Supervision, Investigation.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No. 62476114 and No. 12501640), the Yunnan Fundamental Research Projects (Grant No. 202401AV070009), and the Yunnan Provincial Department of Education Scientific Research Fund Projects (Grant No. 2024J0083).

#### Data availability

Data will be made available on request.

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